



Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE 'A'

Course Code & Title: CS8392 Object Oriented Programming

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP(SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit II / Strings
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** Code Completion

- **Justification:**

Strings is the important concept in java which has many built in methods for processing. In order to remember and recollect how to apply these methods and also to familiarize the syntax, a code completion activity is given to the students.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

Students were asked to form a team of two members for this activity. In this activity, each team was given with a Java program which contains a set of partially completed code and a task to complete the code. The partial code and task are related to the string methods such as split, replace, testing string equality, string transformation.

The activity was conducted during the last 15 minutes of the class. The students have to think and identify the methods to complete the required task in the given code and it is shown in Fig 1. And also, the students have to find the output of the code after completion. After completing the code, every student from a team discussed their code and the findings to all other team members. The completed code snippets are shown in Fig:2-4.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO2	3	3	2	1	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1	PSO3
	3	3	3	2	2	1	1
Justification for correlation	The students apply the	The students evaluate	The students	The students demonstrate	The students discuss	The students apply the	The students use these

	knowledge of strings	given problem statements and identify methods	find the solutions for the given task.	problem-solving skills as a team.	the solutions with others.	string concepts to solve problems	concepts in product design
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• **Images / Screenshot of the practice:**



Fig 1: Student's involvement

Course code & Title: CS8392 Object Oriented Programming
Innovative Practice- Code Completion

Topic: Strings

Complete the given partial code based on the requirements and find the output:

```

public class stringExample{
    public static void main(String args[]){
        String s1="Welcome to Java programming";
        int index2= s1.indexOf("Java") ;//returns the index of Java substring
        System.out.println(index2);
        int index4= s1.lastIndexOf('m') ;//find the last index of m char value
        int index4= s1.lastIndexOf('e', 5) ;//find the last index of e char value from 5th position
        System.out.println(index4);
    }
}

```

Output:

11
6

R. Diwya Dharshini
A. Elavarasi

Fig 2: Sample Code completion sheet

Course code & Title: CS8392 Object Oriented Programming Innovative Practice- Code Completion Topic: Strings Complete the given partial code based on the requirements and find the output: <pre> public class StringTrimExample2 { public static void main(String[] args) { String s1 = " hello java string "; System.out.println(<u>s.length()</u>); // Display the length of the string System.out.println(s1); //Without trim() String tr = <u>s1.trim()</u>; // Apply trim method System.out.println(<u>"length: " + tr.length()</u>); //Display the length of the trimmed string System.out.println(tr); } } </pre> Output: length: 20 length: 17	953619106013 953619106025
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RAMCO INSTITUTE OF TECHNOLOGY Department of Computer Science and Engineering Academic Year: 2021-2022 (Odd Semester) Branch & Semester: V Semester B.E. ECE 'A' Course code & Title: CS8392 Object Oriented Programming Name of the Faculty Member: Mrs.B.Vijayalakshmi Innovative Practice- Code Completion Topic: Strings Complete the given partial code based on the requirements and find the output: <pre> public class stringExample{ public static void main(String args[]){ String s1="This is index of example"; int index1= <u>s1.index(is)</u> ;//find the index of is substring is System.out.println(<u>index 1</u>); // Display the index value int index3= <u>s3.index(is, 4)</u> ;//find the index of is substring after 4th index System.out.println(index3); int index4= <u>s4.index('e', 19)</u> ; //find the last index of e char value System.out.println(index4); } } </pre> Output: ?		U. Harini
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Fig 3: Sample Code completion sheet

M. Aravind
P. Kalidas

RAMCO INSTITUTE OF TECHNOLOGY
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Innovative Practice- Code Completion

Topic: Strings

Complete the given partial code based on the requirements and find the output:

```

public class StringTest {
    public static void main(String[] args) {
        String s1 = new String("Welcome");
        String s2 = new String("WELCOME");
        System.out.println(s1.equals(s2)); // Method to check string equality
        System.out.println(s1.equalsIgnoreCase(s2)); // To check the strings without
        considering case
        String s3 = s2; // To copy the string in s2 to string variable s3
        System.out.println(s1.compareTo(s3)); // To compare the strings s1, s3
    }
}
  
```

Output: False
True
Positive

953619106032
953619106033

Course code & Title: CS8392 Object Oriented Programming
Innovative Practice- Code Completion

Topic: Strings

Complete the given partial code based on the requirements and find the output:

```

public class SplitExample2{
    public static void main(String args[]){
        String s1="Welcome,to,Programming,C, Java, C++, Python";
        System.out.println("Splitted words:");
        String sarr[] = s1.split(","); // Split the string s1 based on comma(,)
        for(String w:sarr){
            System.out.println(w);
        }
        String sarr[] = s1.split(" "); // Split the string s1 into 3 words
        for(String w:sarr){
            System.out.println(w);
        }
    }
}
  
```

Output: welcome
to
programming
c
java
c++

welcome
to
programming, c, java, c++, python ✓

Fig 4:Sample Code completion sheet

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

The completed code was corrected during the hour itself and the following observations were made.

- Some of the students find very easy to identify the method to apply and complete the code easily.
- Some of them need more clarification on the syntax of various string methods.
- Additional examples shall be given them to make understand the concept easily.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

This activity was very useful to identify whether the students understand the string methods. Through this activity, the students are able to identify the places where the string methods can be applied and also the familiarize with the syntax.

- ❖ ***Challenges faced in implementation:***

- Some of the students feel difficult to find out which string methods will complete the given task.
- Some of them feel difficult to follow the syntax of string methods since they were not familiar with the syntax.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/26_OIT551_Colloborativecoding.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/18_EC8393_Codedebug.pdf



Department of Computer Science and Engineering
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Innovative Practice Description

- **Unit / Topic: Unit II / Strings**

- **Course Outcome: CO2**

- **Topic Learning Outcome: TLO5**

- **Activity Chosen: Code Completion**

- **Justification:**

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- Students were asked to form a team of two members for this activity.
- In this activity, each team was given with a Java program which contains a set of partially completed code and a task to complete the code. The partial code and task are related to the string methods such as split, replace, testing string equality, string transformation which is shown in Fig 1- 2.
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- In the first 10 minutes the students have to think and identify the methods to complete the required task in the given code. And also, the students have to find the output of the code after completion.
- After completing the code, during the last 5 minutes every student from a team discussed their code and the findings to all other team members.

- **CO – PO / PSO mapping:**

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Justification for correlation	The students apply the knowledge	The students evaluate given	The students find the	The students demonstrate problem-	The students discuss the	The students apply the string	The students use these concepts

	of strings	problem statements and identify methods	solutions for the given task.	solving skills as a team.	solutions with others.	concepts to solve problems	in product design
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- **Images / Screenshot of the practice:**

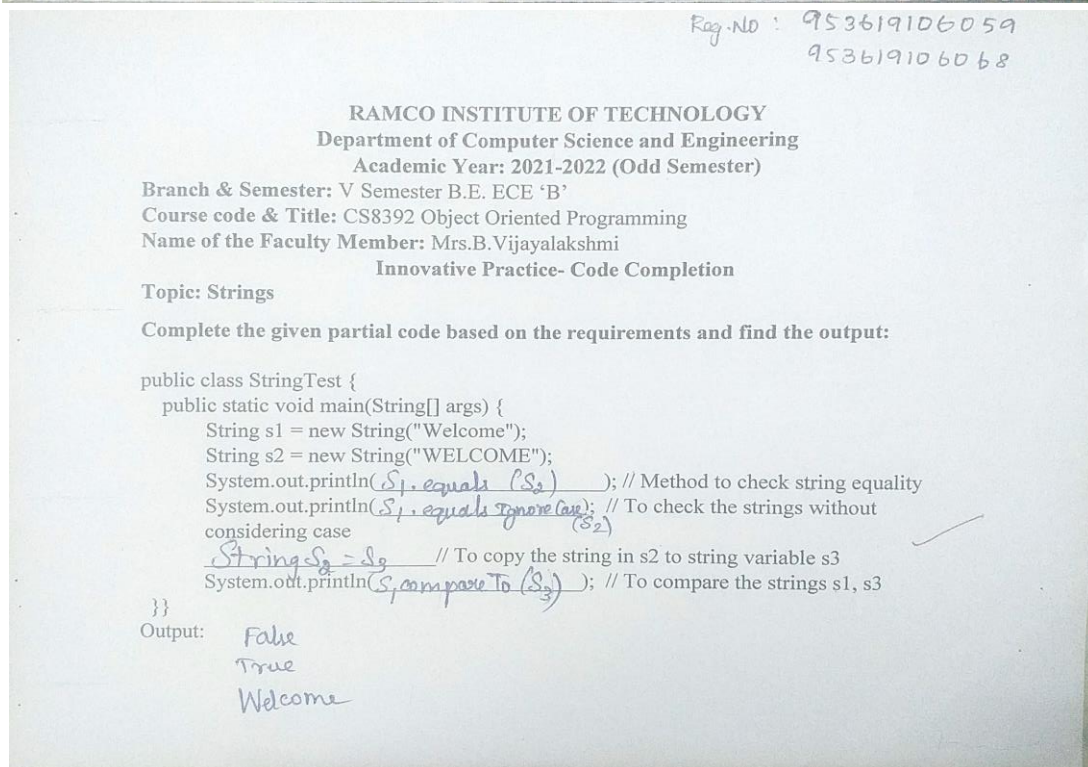
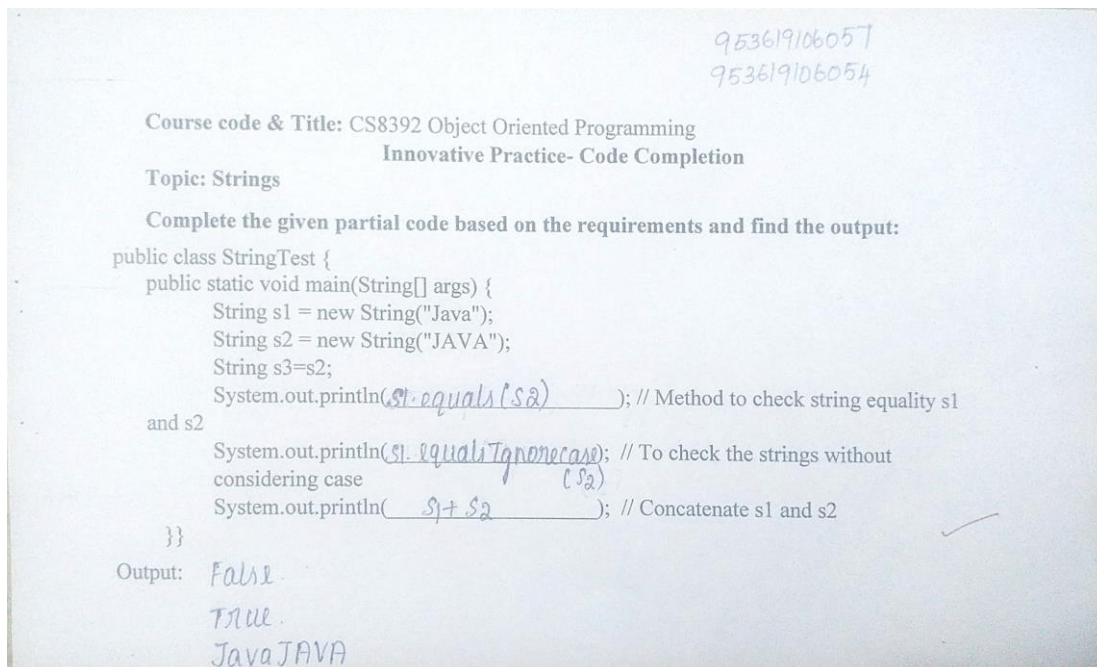


Fig 1: Sample Coding sheet

Course code & Title: CS8392 Object Oriented Programming
Innovative Practice- Code Completion

Topic: Strings

Complete the given partial code based on the requirements and find the output:

```

public class ConcatExample{
    public static void main(String args[]){
        String s1="java string";
        String s2="is immutable";
        String s3="It cannot be changed";
        System.out.println(s1.concat(s2)); //Concatenate s1 and s2
        System.out.println(s1+s2+s3); //Concatenate s1,s2 and s3
    }
}

```

Output:

java string is immutable
java string is immutable It cannot be changed

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Innovative Practice- Code Completion

Topic: Strings

Complete the given partial code based on the requirements and find the output:

```

class CountTheWords
{
    public static void main(String[] args)
    {
        System.out.println("Enter the string");
        Scanner sc = new Scanner(System.in); sc.nextLine(); ✓
        String s= sc.nextLine(); // Read string from user
        String[] words = s.split(" "); // divide the string into words
        System.out.println("Number of words in the string = words.length");
        //Count the number of words in the string array
    }
}

```

Output:

4 words

sc.nextLine();

Fig 2: Sample Coding sheet

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The completed code was corrected during the hour itself and the following observations were made.

- Some of the students find very easy to identify the method to apply and complete the code easily.
 - Some of them need more clarification on the syntax of various string methods.
 - Additional examples shall be given them to make understand the concept easily.
- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 This activity was very useful to identify whether the students understand the string methods. Through this activity, the students are able to identify the places where the string methods can be applied and also the familiarize with the syntax.
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